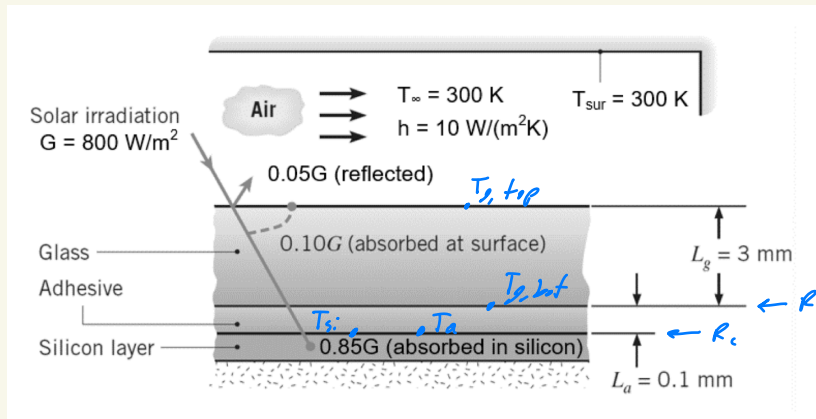


#1

→ Known: $k_g, k_a, \eta = a - bT_{si}$, a, b , back side insulated,
 R_c, b , absorption %'s, $T_\infty, h, T_{sur}, h_{rad}$, SB law unnecessary

→ Schematic:



→ Find:

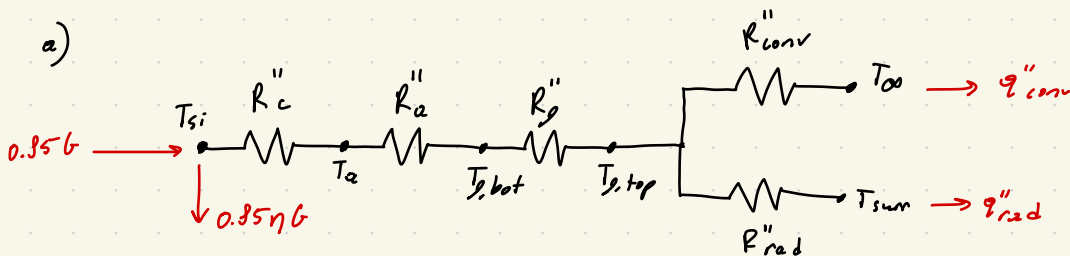
- Thermal resistance network
- $T_{g,top}$ & T_{si}
- η & Power

→ Properties: N/A

→ Assumptions: perfect insulation, no thermal resistance between glass and adhesive, SS

→ Analysis: use R'' 's bc they all have same area

a)



R'' type	Value [$m^2 K/W$]
R''_c	10^{-4}
R''_a	10^{-4}
R''_g	0.002
R''_{conv}	0.1
R''_{rad}	0.2

Key assumptions: 1D
SS

$$q'' = \frac{\Delta T}{R}$$

b) Silicon energy balance: $q'' = \frac{\Delta T}{R''_{tot}}$

$$0.856 - \underbrace{0.85\eta G}_{\text{"lost" to electricity generation}} = \frac{T_{si} - T_{g,top}}{R''_c + R''_a + R''_g} \Rightarrow 0.856(\eta - 1) = \frac{T_{si} - T_{g,top}}{R''_c + R''_a + R''_g}$$

$$304 + 0.68T_{si} = \frac{T_{si} - T_{g,top}}{2.2 \times 10^{-3}}$$

need another eq'n

Energy balance on glass: $856(1-\eta)$ absorption from silicon conducts to glass

$$0.856(1-\eta) + 0.16 = \frac{T_{g, top} - T_{\infty}}{R''_{conv}} + \frac{T_{g, top} - T_{sur}}{R''_{rad}}$$

$$680(0.447 + 0.001T_{si}) + 80 = \frac{T_{g, top} - 300}{0.1} + \frac{T_{g, top} - 300}{0.2}$$

Trusty computer $\Rightarrow$ $T_{si} = 342.3 \text{ K}$ $T_{g, top} = 341.1 \text{ K}$

c) $\eta = a - bT_{si} = 0.553 - 0.001(342.3) = 0.21$

Power per area $= 0.856\eta = 0.85 \cdot 800 \frac{\text{W}}{\text{m}^2} \cdot 0.21 = 143 \text{ W/m}^2$